

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-24

Faculty

Dr. Rajasekhar N V

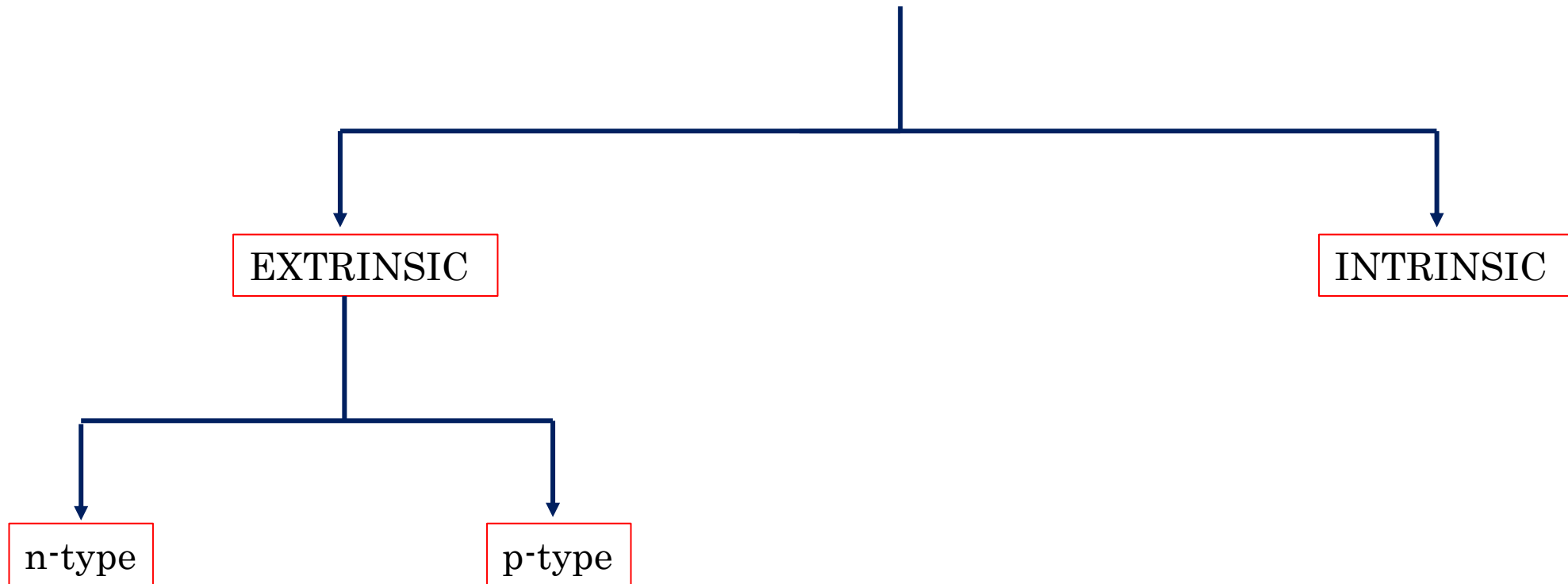
Associate Professor G-1

SENSE, VIT-AP University

Mail: rajasekhar.venkata@vitap.ac.in

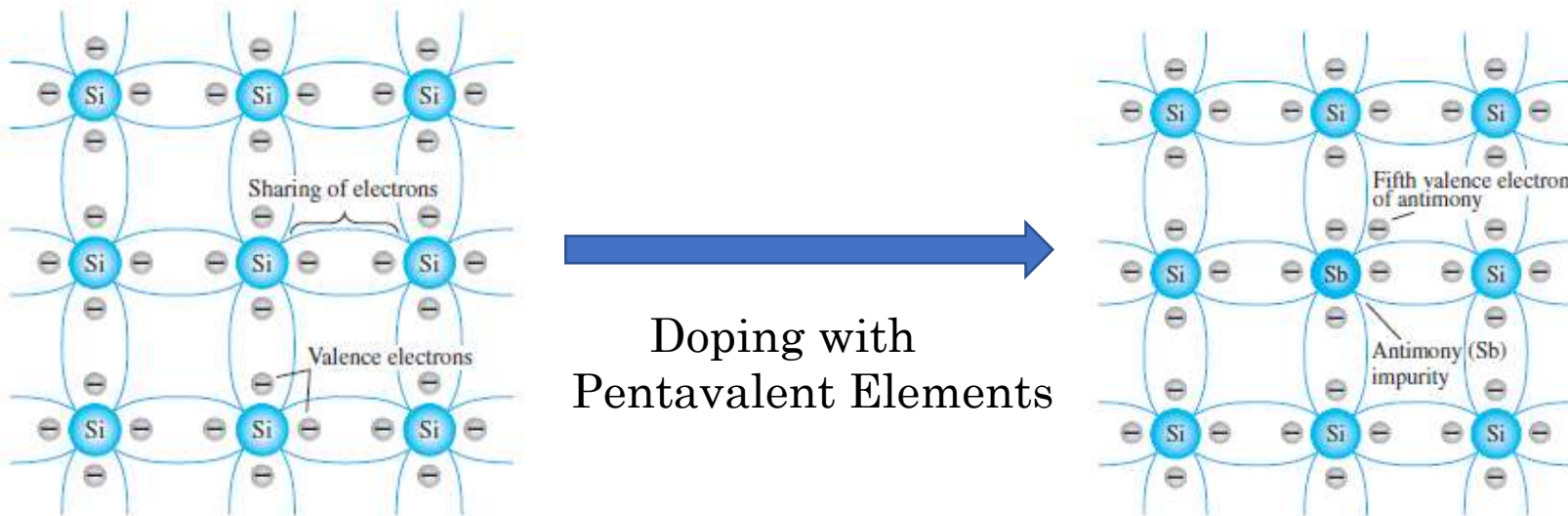
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Semiconductor



1.n-type Semiconductor

- n -type materials are formed by adding a predetermined number of impurity atoms to a silicon base.
- An n -type material is created by introducing impurity elements that have five valence electrons (pentavalent), such as antimony , arsenic , and phosphorus.
- Diffused impurities with five valence electrons are called donor atoms.

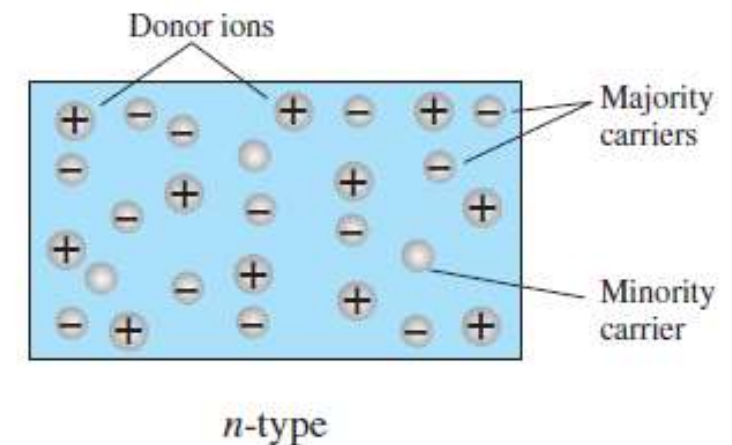
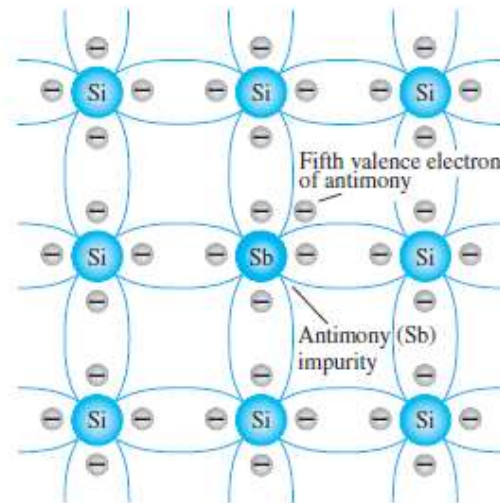


- In an n-type material the electron is called the majority carrier and the hole is the minority carrier.

Antimony impurity in n-type material.

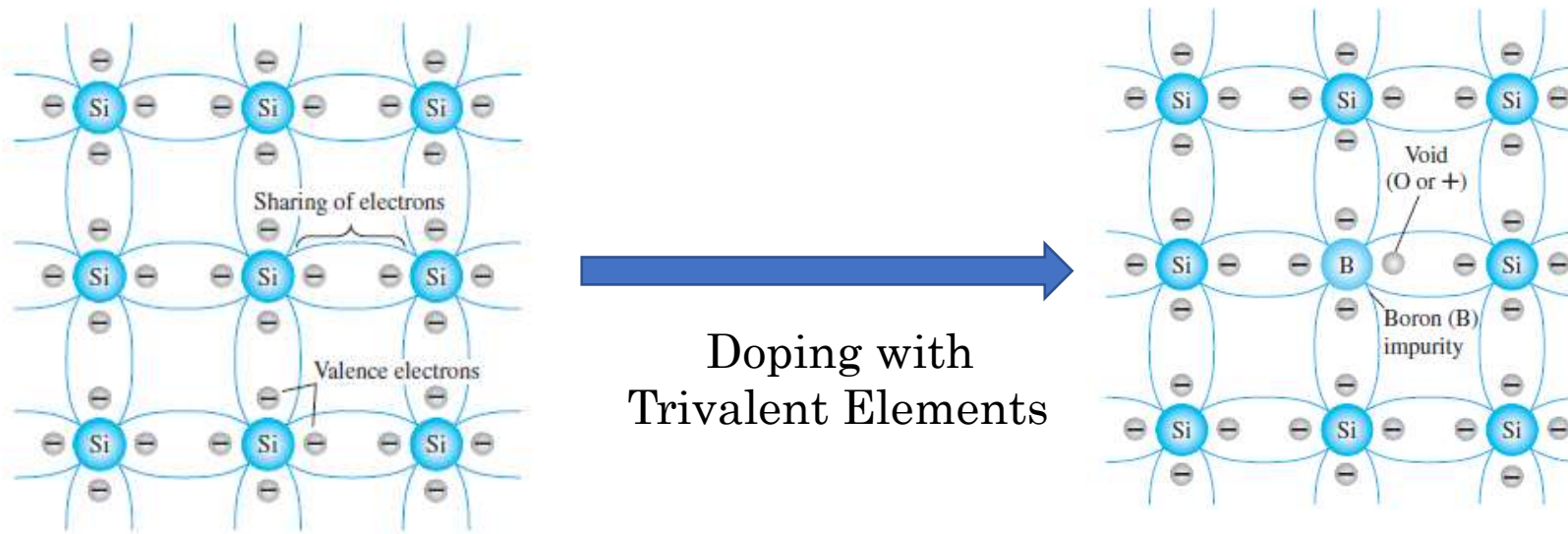
Majority and Minority Carriers

- In an n-type material, the number of holes has not changed significantly from this intrinsic level. The net result, therefore, is that the number of electrons far outweighs the number of holes.
- For this reason: In an n-type material the electron is called the majority carrier and the hole the minority carrier.
- When the fifth electron of a donor atom leaves the parent atom, the atom remaining acquires a net positive charge: hence the plus sign in the donor-ion representation. For similar reasons, the minus sign appears in the acceptor ion.



2. p-type Semiconductor

- The p-type material is formed by doping a pure germanium or silicon crystal with impurity atoms having three valence electrons. The elements most frequently used for this purpose are boron, gallium, and indium.
- The diffused impurities with three valence electrons are called acceptor atoms.

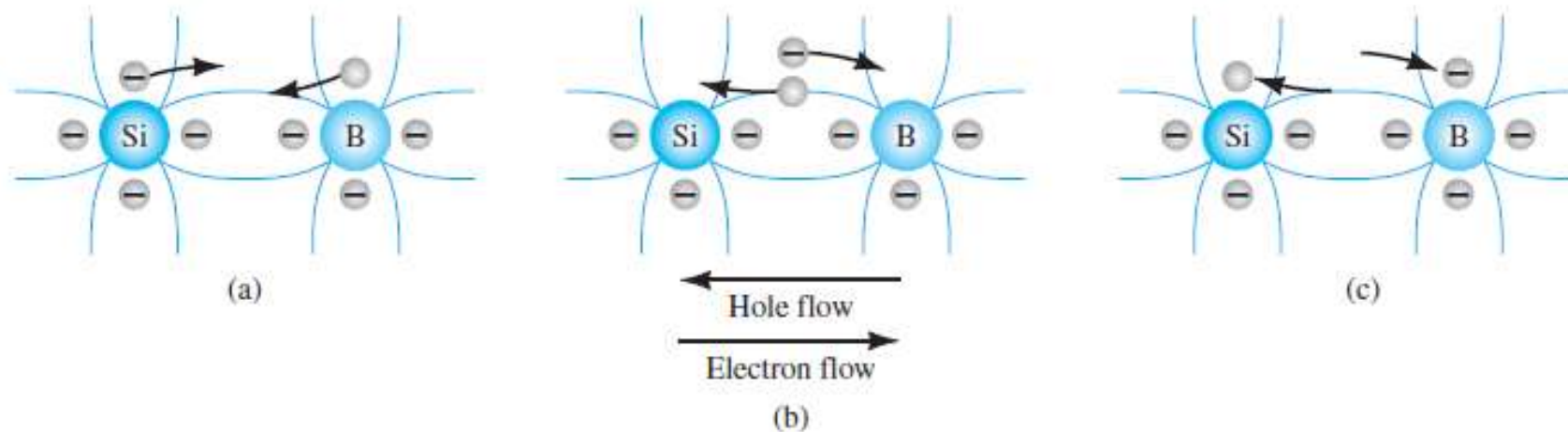


Boron impurity in p-type material.

- In an p-type material the hole is called the majority carrier and the electron is the minority carrier.

p-type Semiconductor

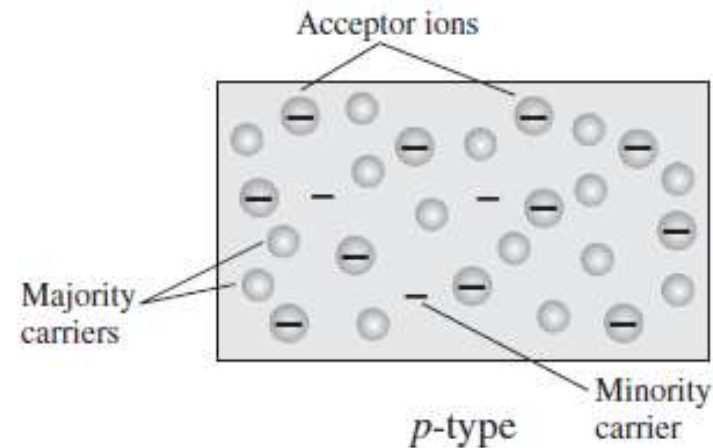
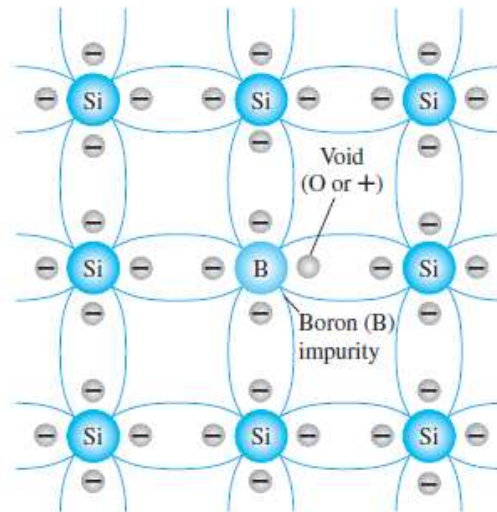
- If a valence electron acquires sufficient kinetic energy to break its covalent bond and fills the void created by a hole, then a vacancy, or hole, will be created in the covalent bond that released the electron. There is, therefore, a transfer of holes to the left and electrons to the right, as shown in Fig. below .
- The direction to be used in this text is that of conventional flow , which is indicated by the direction of hole flow.



Electron versus hole flow.

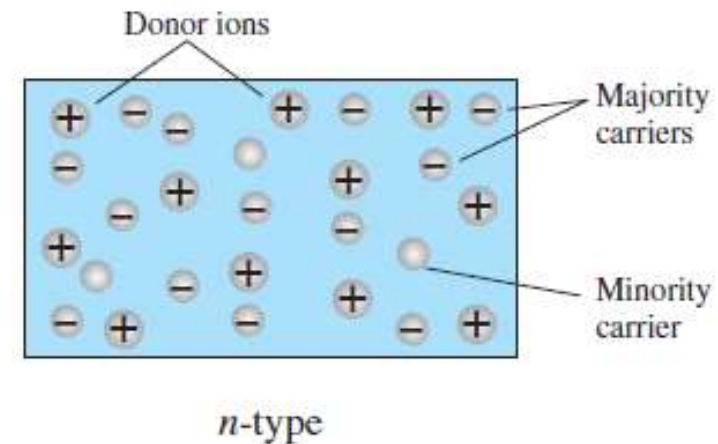
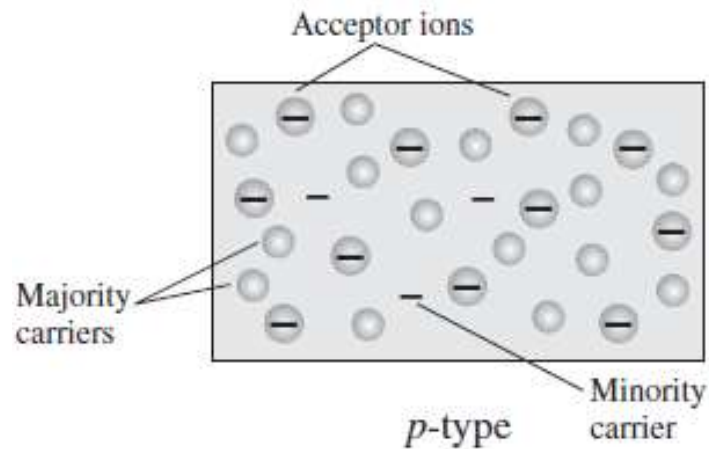
Majority and Minority Carriers

- For the p -type material the number of holes far outweighs the number of electrons. In a p-type material the hole is the majority carrier and the electron is the minority carrier.
- When the acceptor atoms gains electrons from other atoms and the atom remaining acquires a net negative charge: hence, the minus sign appears in the acceptor ion.

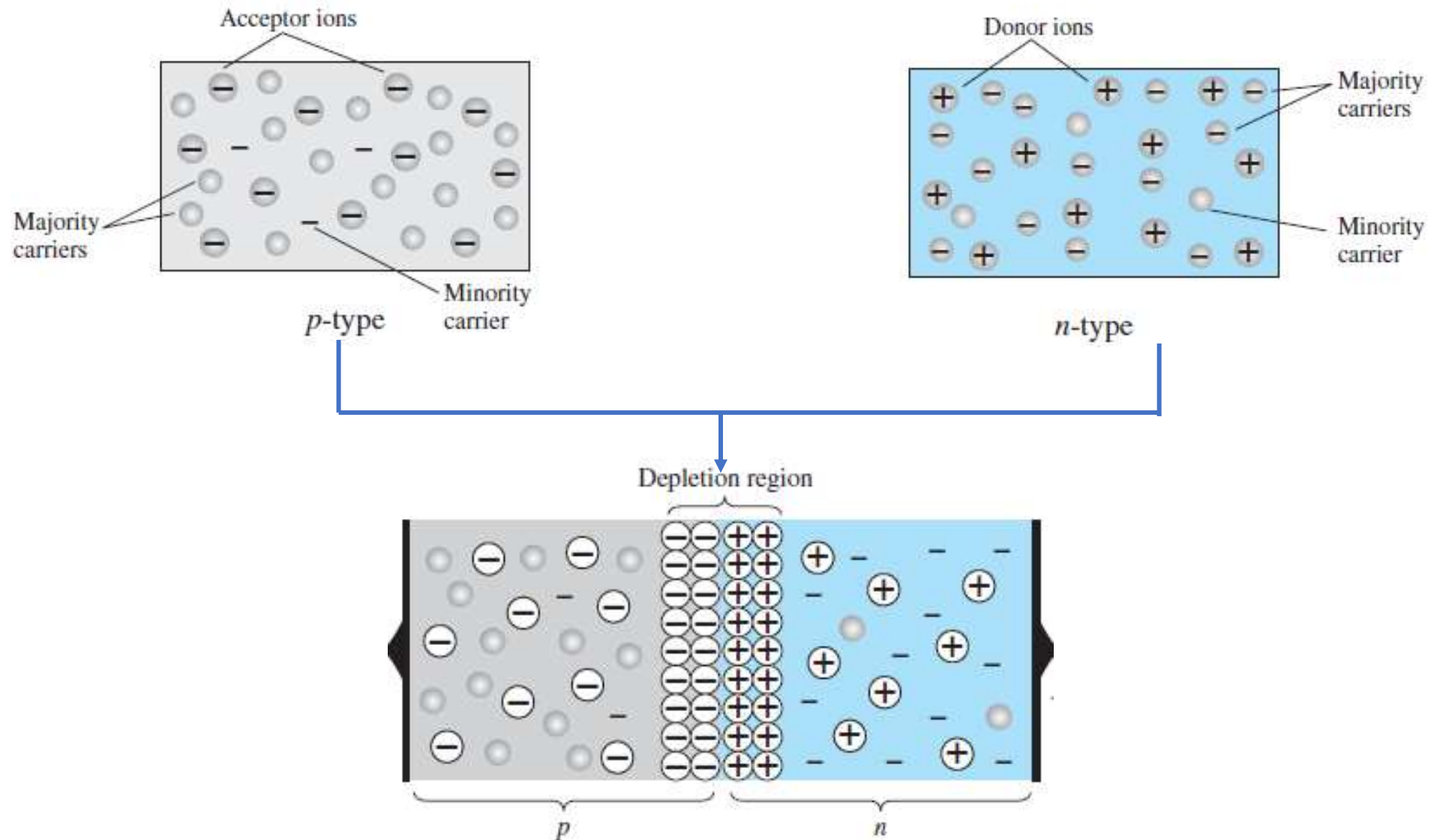


Semiconductor Diode:

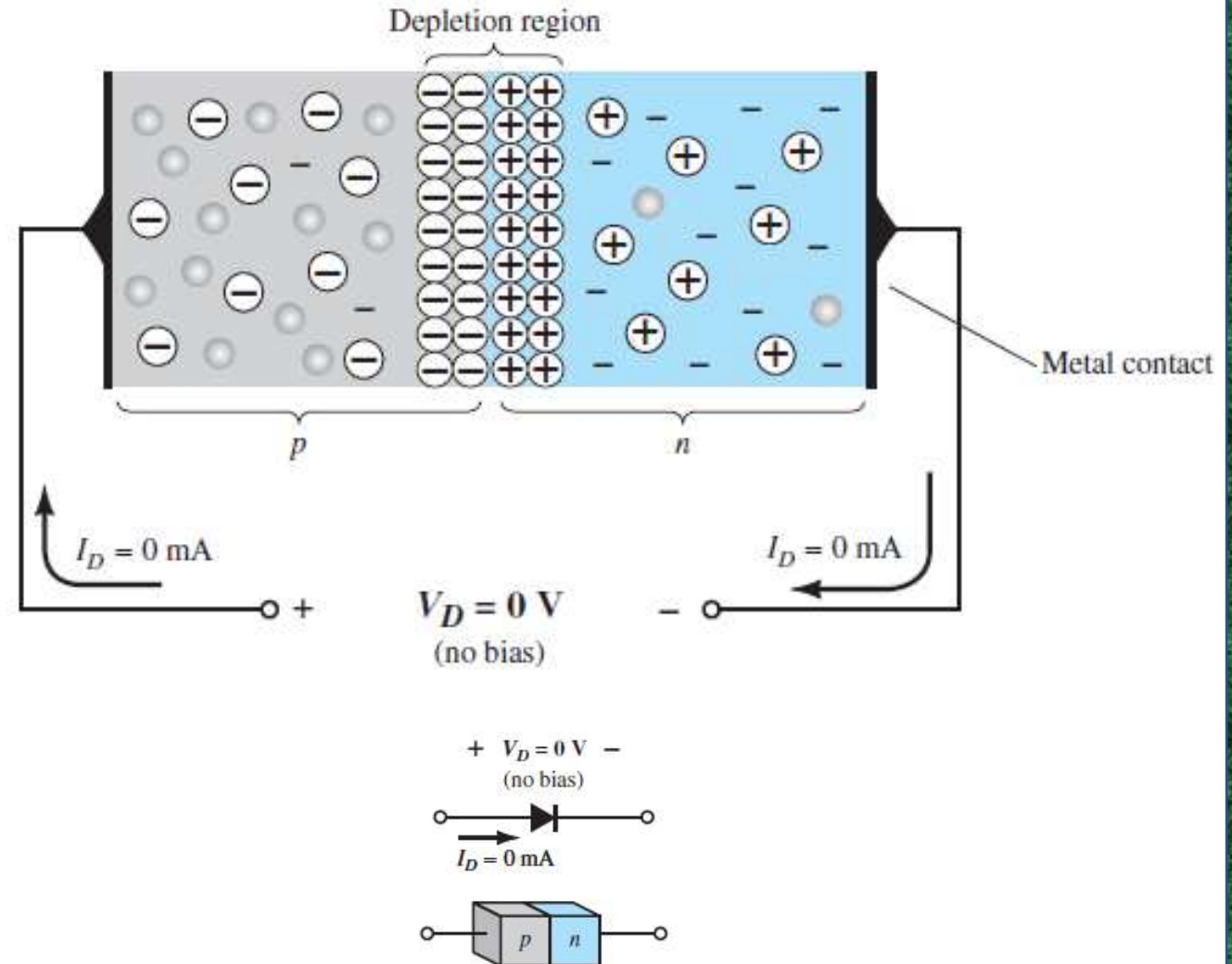
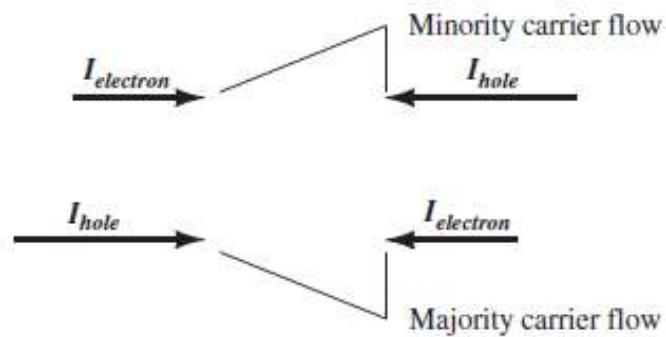
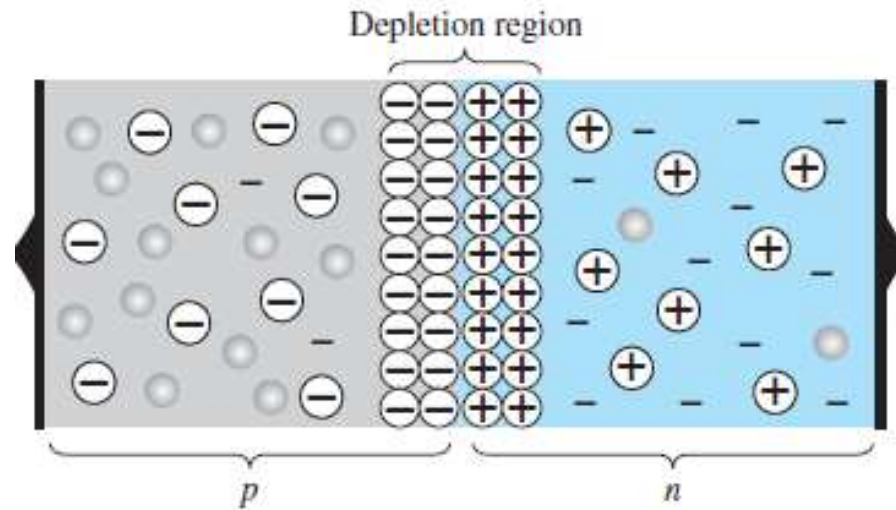
- The semiconductor diode, is created by simply joining a **n-type** and a **p-type** material together, nothing more, just the joining of one material with a majority carrier of electrons to one with a majority carrier of holes.



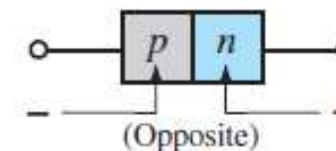
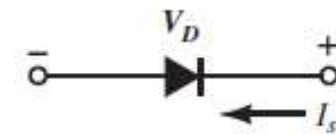
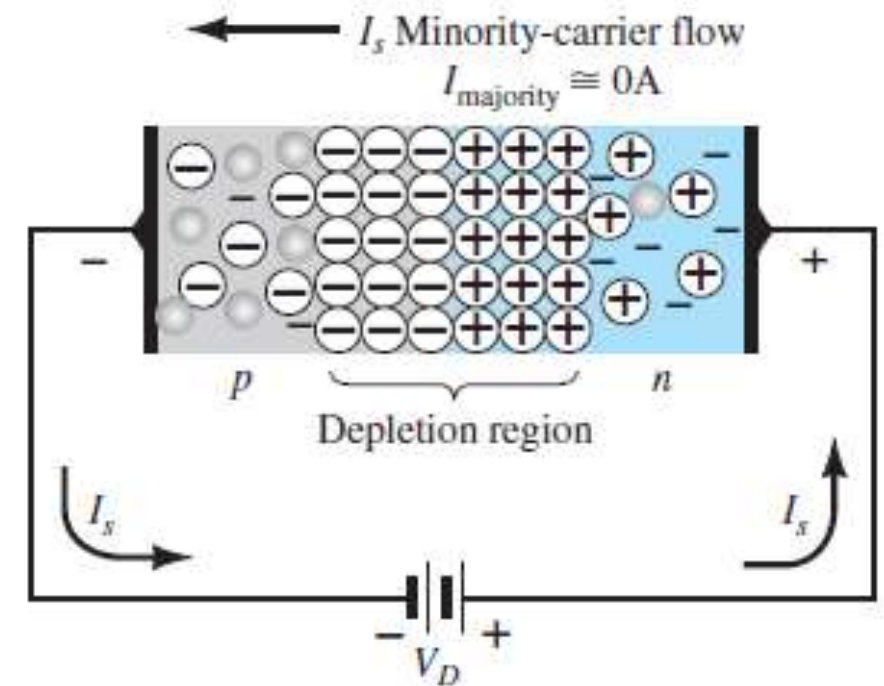
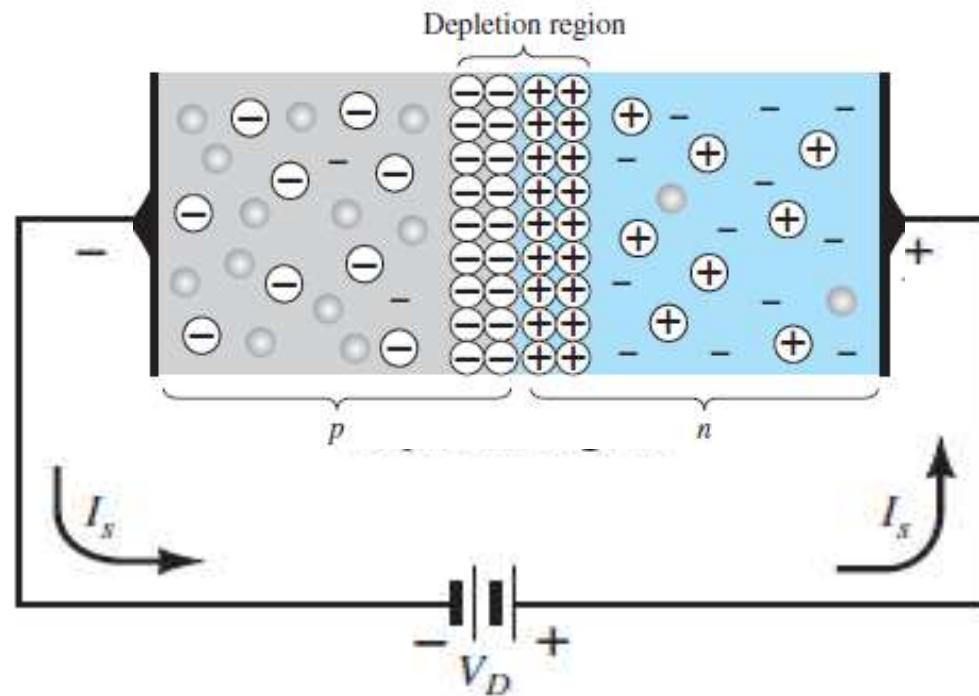
1) No Applied Bias ($V=0$ V)



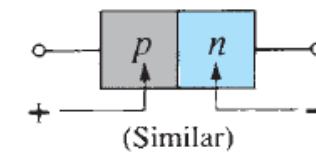
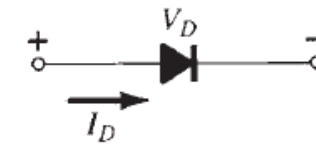
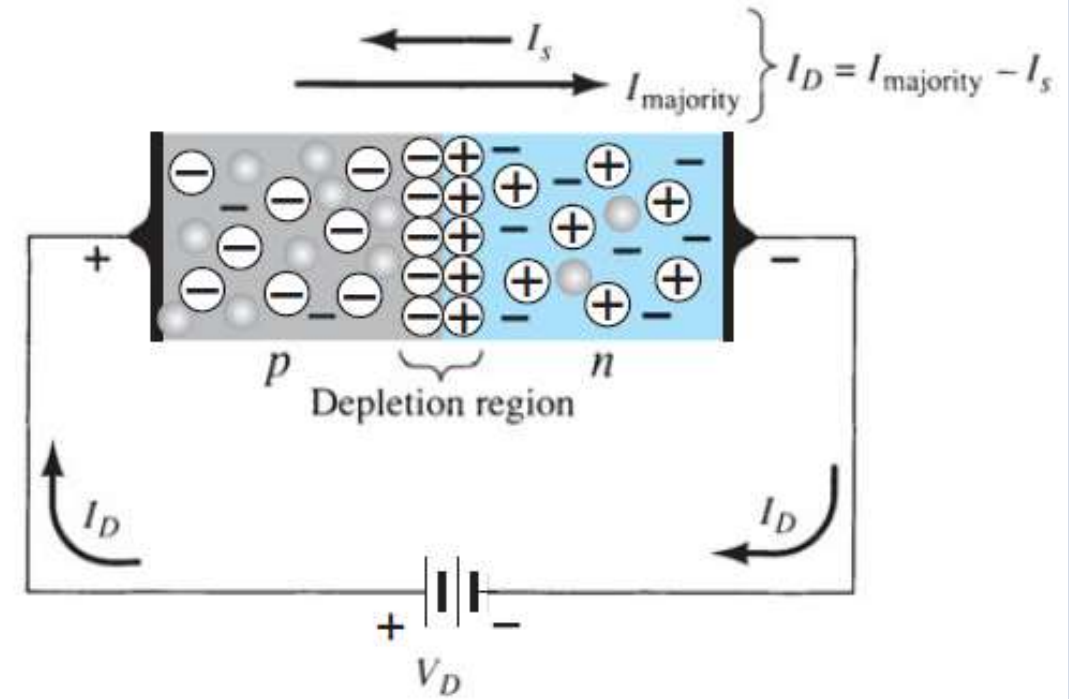
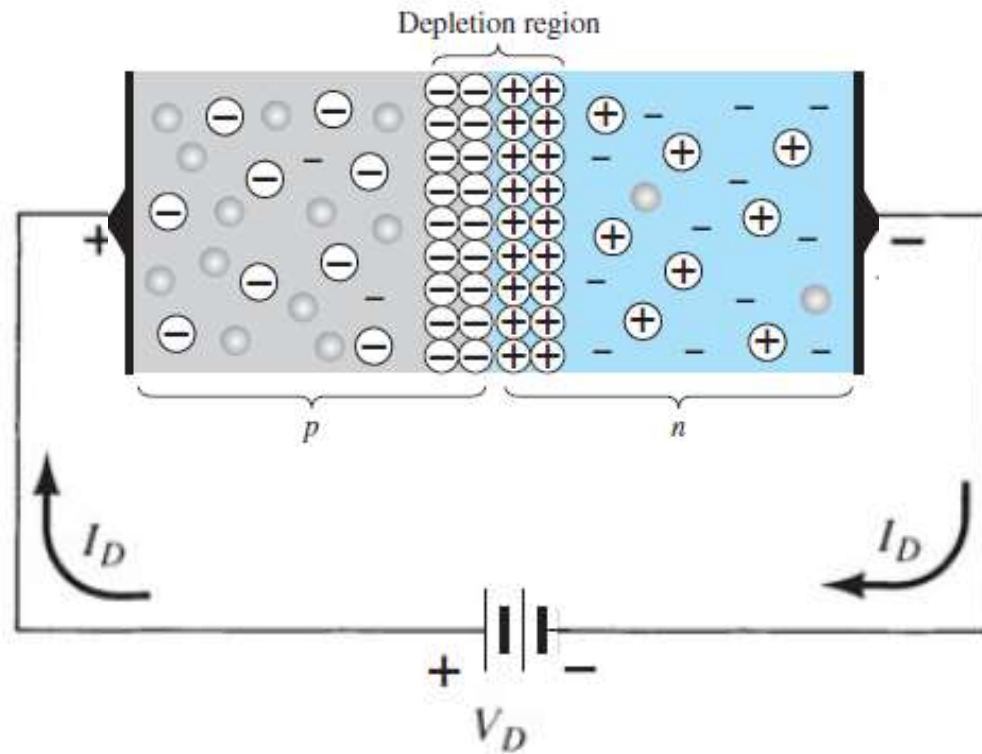
1) No Applied Bias ($V=0$ V)



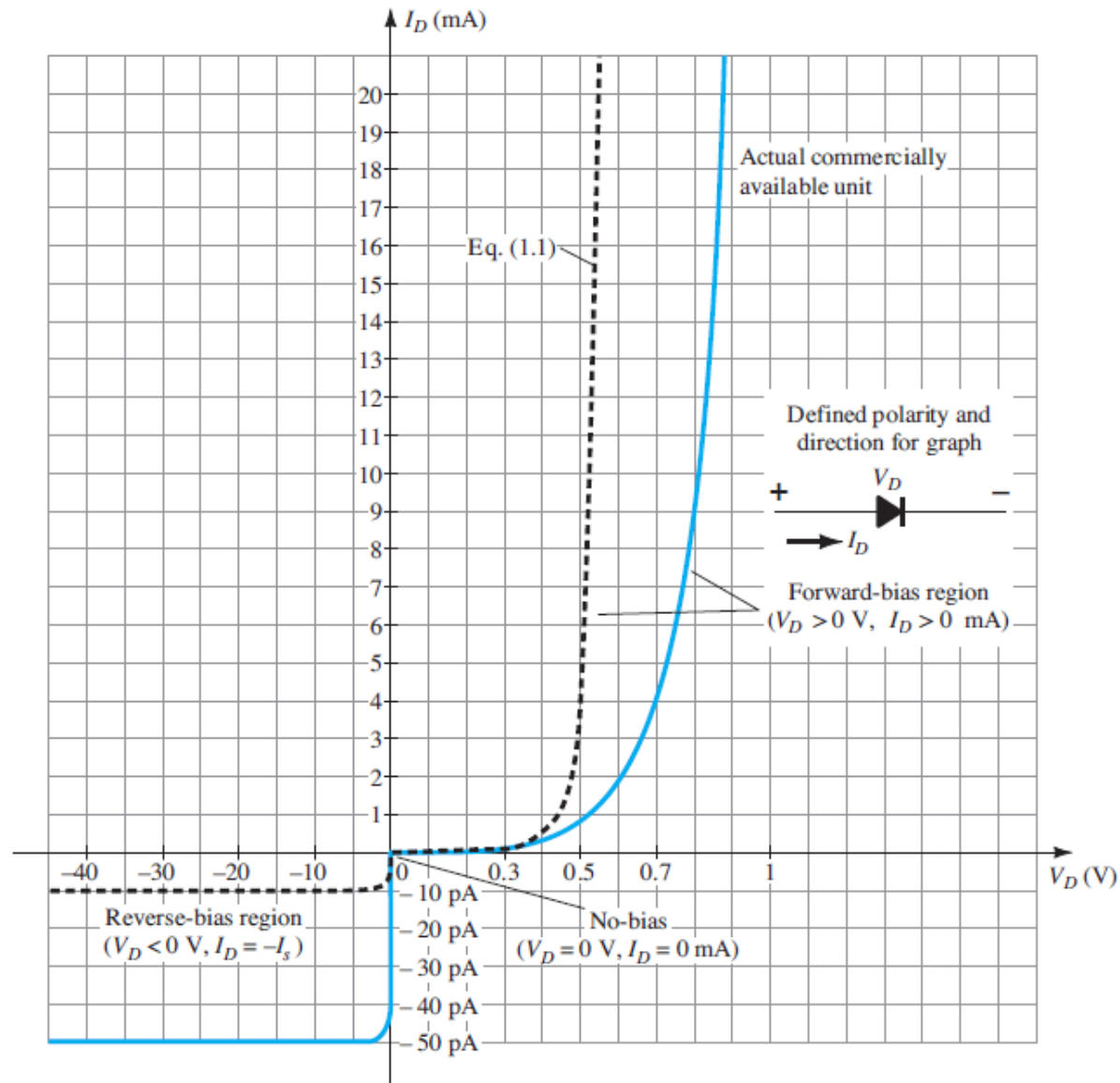
2) Reverse-Bias Condition ($V_D < 0$ V)



3) Forward-Bias Condition ($V_D > 0$ V)



Silicon semiconductor diode characteristics.



$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

- I_s = reverse saturation current
- V_D = applied forward-bias voltage across the diode
- n = an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ($n = 1$ will be assumed default value unless otherwise noted).
- V_T = Thermal voltage

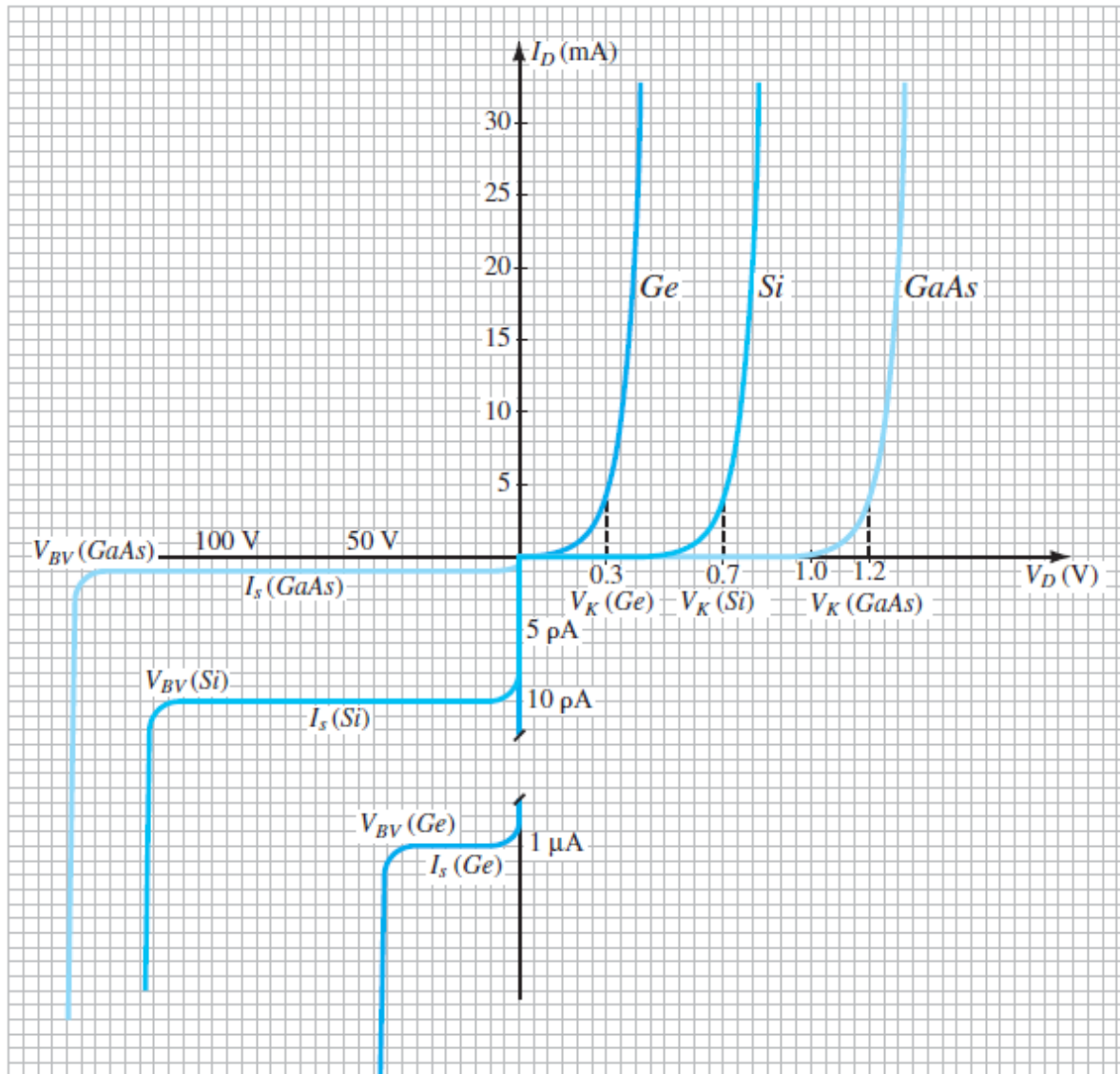
$$V_T = \frac{kT_K}{q} \quad (\text{V})$$

k is Boltzmann's constant = 1.38×10^{-23} J/K

T_K is the absolute temperature in kelvins = $273 +$ the temperature in $^{\circ}\text{C}$

q is the magnitude of electronic charge = 1.6×10^{-19} C

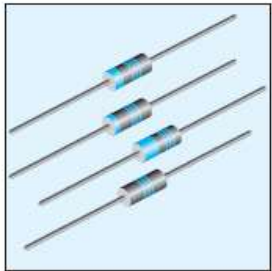
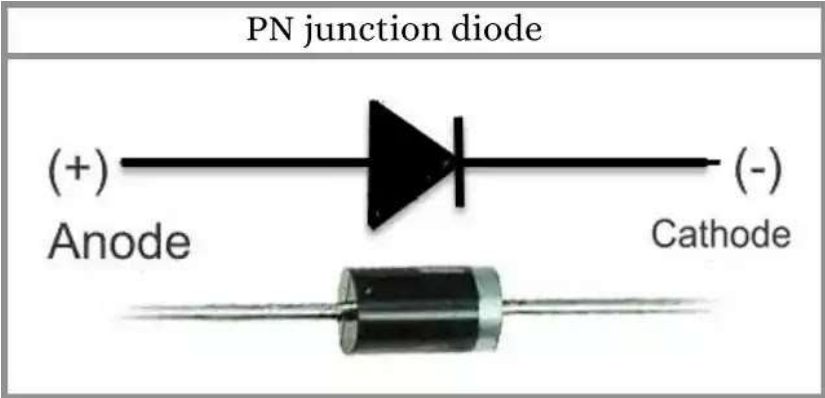
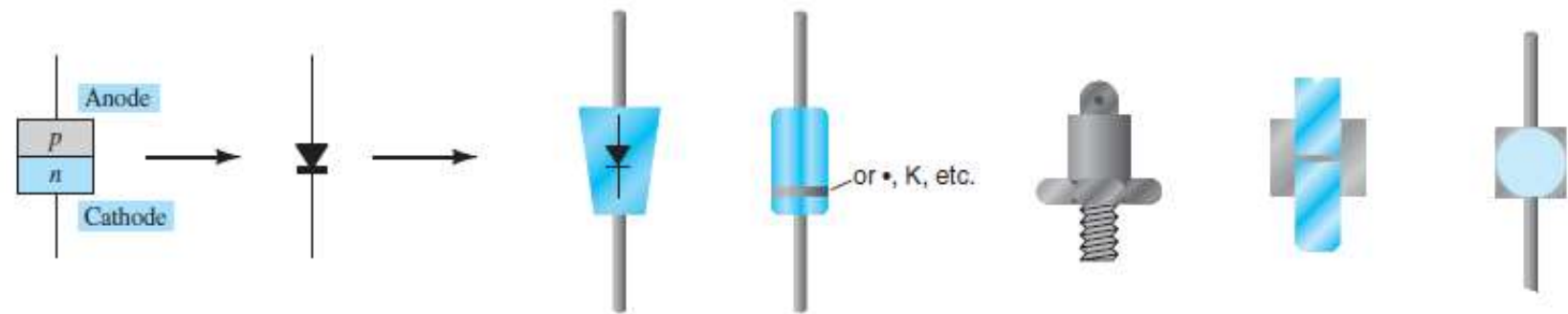
Comparison of Ge, Si, and GaAs commercial diodes.



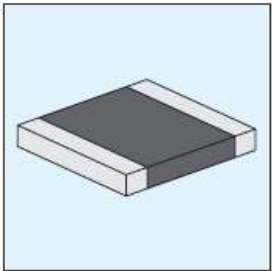
Knee Voltages V_K

Semiconductor	V_K (V)
Ge	0.3
Si	0.7
GaAs	1.2

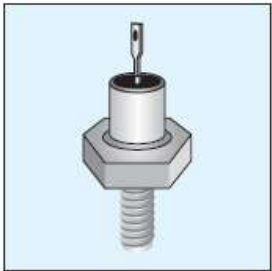
Semiconductor Diode Notation



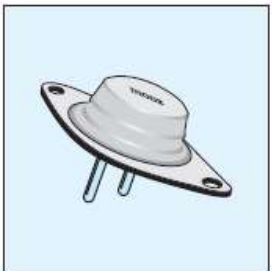
General purpose diode



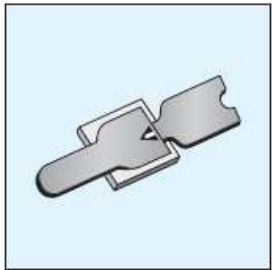
Surface mount high-power PIN diode



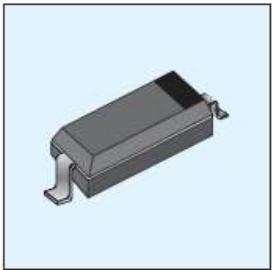
Power (stud) diode



Power (planar) diode



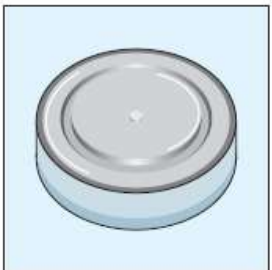
Beam lead pin diode



Flat chip surface mount diode



Power diode



Power (disc, puck) diode

1. At a temperature of 27°C (common temperature for components in an enclosed operating system), determine the thermal voltage V_T .

$$T = 273 + ^\circ\text{C} = 273 + 27 = 300 \text{ K}$$

$$\begin{aligned} V_T &= \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})}{1.6 \times 10^{-19} \text{ C}} \\ &= 25.875 \text{ mV} \cong 26 \text{ mV} \end{aligned}$$

2. (a) Determine the thermal voltage for a diode at a temperature of 20°C. (b) For the same diode of part (a), Find the diode current if $I_s = 40 \text{ nA}$, $n = 2$ (low value of V_D), and the applied bias voltage is 0.5 V.

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$V_T = \frac{kT_K}{q} \quad (\text{V})$$

k is Boltzmann's constant $= 1.38 \times 10^{-23} \text{ J/K}$

T_K is the absolute temperature in kelvins $= 273 + \text{the temperature in } ^\circ\text{C}$

q is the magnitude of electronic charge $= 1.6 \times 10^{-19} \text{ C}$

3. Determine the diode current at 20°C for a silicon diode with $n=2$, $I_s = 0.1 \mu\text{A}$ at a reverse-bias potential of -10 V.

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$V_T = \frac{kT_K}{q} \quad (\text{V})$$

4. Given a diode current of 8 mA and $n = 1$, find I_s if the applied voltage is 0.5 V and the temperature is room temperature (25°C).

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$V_T = \frac{kT_K}{q} \quad (\text{V})$$

5. Given a diode current of 6 mA, $V_T = 26$ mV, $n = 1$, and $I_s = 1$ nA, find the applied voltage V_D .

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$V_T = \frac{kT_K}{q} \quad (\text{V})$$